

Er. HILLOL DAS

AIoT Researcher | Cloud-based Distributed System Integrator | Academic Expert | App Developer

hillol.das@gov.in | +91 9774840986 / 7005917758 | <https://www.linkedin.com/in/hillolishere>

PROFESSIONAL OBJECTIVE

A passionate individual who always thrives to work on end to end products which develop sustainable and scalable social and technical systems to create impact. Self-driven, quick starter, passionate programmer with a curious mind who enjoys solving a complex and challenging real-world problems. To acquire an opportunity to establish an academic career in an organization where I can do meaningful works for the development of my teaching ability and which offers opportunity for growth, enriching the experience and to achieve excellence in the field of my work.

AREAS OF INTEREST

- e-Governance & Consulting: Digital public infrastructure and capacity building.
- Geo-Spatial Analytics: GIS and remote sensing for rural planning.
- AIoT, Cloud & HCI: Smart automation, serverless architecture.
- AI, ML & Vision: Predictive modeling, data algorithms, and pattern recognition.
- Mobile & Software Development: End-to-end application architecture and deployment.

PROFESSIONAL EXPERIENCE

Faculty (Computer)

Department of RD (Panchayat), Govt. of Tripura (28th June 2024 - Present)

Panchayat Resource Development Officer (Computer)

Department of RD (Panchayat), Govt. of Tripura (22nd January 2016 - 28th June 2024)

District eGovernance Manager

NIELIT Agartala, Govt. of India (10th November 2014 - 21st January 2016)

Visiting Lecturer

Bir Bikram Memorial College, Agartala, India (July 2014 - November 2014)

Assistant Professor

Holy Cross College, Agartala, India (1st July 2014 - 31st December 2014)

Teaching Assistant

Department of Computer Science & Engineering, NIT Agartala (under GATE scholarship) (August 2012 - June 2014)

Technical Assistant

Department of Computer Science & Engineering, NIT Agartala (March 2012 - July 2012)

Computer Instructor

PARAM Super Computer Lab, C-DAC Satellite Center, NIT Agartala (August 2011 - February 2012)

ACADEMIC PROFILE

Master of Technology in Computer Science and Engineering - National Institute of Technology, Agartala, Government of India (2014)

Post Graduate – Diploma in Advanced Computing - Centre for Development of Advanced Computing (C-DAC) ACTS, PUNE (2012)

Bachelor of Engineering in Computer Science and Engineering - Tripura Institute of Technology, Agartala, Tripura (2011)

All India Senior School Certificate Examination (AISSCE) - Jawahar Navodaya Vidyalaya Hailakandi School, Assam (2007)

All India School Certificate Examination (AISCE) - Jawahar Navodaya Vidyalaya Hailakandi School, Assam (2005)

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Dart, C++, Java, C#

Frameworks: Flutter, React.js, Node.js, Spring Boot, Next.js

Databases: PostgreSQL, MongoDB, MySQL, Redis, Firebase Firestore

Tools & Platforms: LLM, Computer Vision, Microsoft Azure, AWS, Docker, Git, AI Studio, VS Code, Android Studio, IntelliJ IDEA, Jupyter Notebook, Visual Studio, Arduino IDE

KEY PROJECTS

Gram Barta (Audio Broadcast App) (2026) - Android Studio (Kotlin), Firebase Cloud Messaging, Realtime Database, and Serverless Backend Integration

A cloud-integrated platform developed for Bijohnagar Gram Panchayat to digitize rural communication. Architected a real-time broadcast system to enhance e-governance and citizen engagement.

Tripura Calendar (2018) - Android Studio (Kotlin/Java)

A specialized digital utility application featuring state-specific administrative schedules and holidays. Engineered with a focus on localized UI/UX and high-performance accessibility for regional users.

Automatic Detection and Analysis of Retinal Diseases (2014) - Image Processing / Healthcare Tech

Engineered an automated system capable of analyzing fundus images to detect and classify retinal diseases.

Hand Gesture-Based Home Appliance Controller (2013) - Embedded Systems / Automation

Developed an interactive system allowing users to seamlessly control various home appliances through hand gesture recognition.

Research Project: Security in Wireless Sensor Networks (WSN) (2012) - Network Security

Conducted an in-depth analytical study on common security vulnerabilities, attacks, and effective countermeasures within WSNs.

Online Sales & Inventory Management System (2012) - Struts 2, Hibernate Framework, Oracle 11g

Built a robust backend and web application to streamline online sales tracking and inventory management.

E-Resources Platform (2011) - ASP.NET, SQL Server 2005

Designed and implemented a comprehensive digital platform for managing and accessing electronic resources.

Research Project: 'Blue Brain' Artificial Intelligence (2011) - Artificial Intelligence / Neural Simulations

Presented a detailed technical analysis on the EPFL and IBM combined project focusing on simulating the human brain.

Training Portal Site (2010) - J2EE (Servlet, JSP, JavaBeans), Oracle 10g

Developed an interactive web-based portal to facilitate and manage training resources and user access.

Industrial Project & Training in .NET Technologies (2010) - .NET Framework

Completed a one-month intensive, hands-on industrial program focusing on real-world application development using .NET.

PUBLICATIONS

- Hillol Das, Ashim Saha, Suman Deb and Gautam Nath. "An Automated Approach to Differentiate a Normal eye from a Defective Eye". International Journal of Computer Applications 79(15):10-15, October 2013. Published by Foundation of Computer Science, New York, USA.
- Ashim Saha, Hillol Das, Nirmalya Kar and M C Pal. "An Approach of Extracting Facial Components for Facial Expression Detection using Fiducial Point Detection". International Journal of Computer Applications 80(10):49-53, October 2013. Published by Foundation of Computer Science, New York, USA.
- Nikhil Debbarma, Gautam Nath and Hillol Das. "Analysis of Data Quality and Performance Issues in Data Warehousing and Business Intelligence". International Journal of Computer Applications 79(15):20-26, October 2013. Published by Foundation of Computer Science, New York, USA.
- Hillol Das, Ashim Saha, Suman Deb. "An expert system to distinguish a defective eye from a normal eye". 2014 International Conference on Issues & Challenges in Intelligent Computing Techniques (ICICT 2014), IEEE ISBN: 978-1-4799-2899-6 pp159-162.
- Hillol Das, Ashim Saha, Suman Deb. "Locating Optic Disk in Retinal Fundus Images using Normalized Cross Correlation". International Conference on Emerging Research in Computing, Information, Communication and Applications, ERCICA 2014, ELSEVIER, ISBN: 9789351072638, Bangalore, INDIA.
- Er. Hillol Das, "Strategies to make Training Programs more effective at Panchayat Raj Training Institute". International Seminar & 48th Regional Science Conference on "Rural Habitat, Institutions and Development: Changing Nature & Challenges" held from 5th - 7th January, 2017 at Tripura University, India.
- Hillol Das. "Automatic Detection and Analysis of Retinal Diseases from Fundus Images". ISBN: 9783-659-57415-3, Publisher: LAP LAMBERT Academic Publishing, Germany.

TRAININGS & WORKSHOPS

2026

- Attended Management Development Program from 19th Jan to 23rd Jan 2026 at the IIM Shillong.
- Five-day Faculty Development Programme (FDP) from 25th February to 1st March, 2026 at NIRD Hyderabad.

2025

- Two-Day National Level Write-shop on the "Panchayat Advancement Index (PAI) Version 2.0" held on 26th - 27th May 2025 at Dr. Ambedkar International Centre, New Delhi.
- Two Day Training Workshop on "e-Gramswaraj Enhancements and Modifications in Planning, Accounting and Profiler" held during 9th - 10th September 2025 at Dr. Ambedkar International Centre (DAIC), 15 Janpath, New Delhi.
- 3rd Edition of "NESAC User Interaction Meet (NeUIM)-2025" scheduled to be held during 25th - 26th September 2025 via Hybrid mode at NESAC, Dept. of Space, Govt. of India, Meghalaya.
- "Exposure Visit to the State of Karnataka" held from 27th to 28th October 2025.

2024

- "Training of Trainers (ToT) programs on Localization of SDGs for PRIs: Village with Good Governance" held from 29th - 31st January 2024 at NIRD & PR, SIRD Assam, for State Level Master Trainers (SLMTs).
- National Training Program on "Research Methodology: Analysis, Interpretation and Dissemination of Data for RD and CSR Professionals" held from 14th - 16th February 2024 at NIRD&PR, Delhi Branch, India Habitat Centre, New Delhi.

- One Day National Workshop on "Up-grading skills for preparing of Training Calendar" under revamped RGSA held on 19th June 2024 at NIRD&PR, Hyderabad.
- National Panchayat Awards Ceremony held at Vigyan Bhawan, New Delhi on 11th December 2024.
- Five (05) days Refresher Training Programme for preparation of AAP of RGSA and furnishing the QPR held from 9th - 13th December 2024 at IIPA, New Delhi.

2023

- National Workshop on "ISO Certification of Panchayats" held from 5th to 7th July 2023 at Kerala Institute of Local Administration (KILA), Kerala, organized by the Ministry of Panchayati Raj (MoPR), Government of India.
- Two-Day National Level Write-shop on "Panchayat Development Index (PDI)" held from 10th - 11th August 2023 at Dr. Ambedkar International Centre (DAIC), New Delhi, organized by the Ministry of Panchayati Raj (MoPR), Government of India.

2022

- 3-day training on "Village Poverty Reduction Plan (VPRP)" held from 1st to 3rd August 2022 at Kolkata, initiated by the RD (Panchayat) Department, Government of Tripura.
- National Writeshop on revamped "National Panchayat Awards" held from 16th to 18th August 2022 at New Delhi, organized by the Ministry of Panchayati Raj, Government of India.

2021

- One day National Level "Consultative Workshop on Service Delivery by Panchayats" held on 22nd November 2021 at ANSSIRD in Mysuru, organized by NIRD & PR, Hyderabad.

2020

- Training on "Blockchain Technology" held on 14th January, 2020 at Pragna Bhaban, Agartala, India organized by DIT, Agartala, Government of Tripura in collaboration with National Institute of Smart Government.
- Training on "Digital Governance" held on 28th January, 2020 at Pragna Bhaban, Agartala, India organized by DIT, Agartala, Government of Tripura in collaboration with National Institute of Smart Government.
- "Trainers of Training Programme on Direct Trainers Skill (DTS)" held from 17th - 21st March 2020 at SIPARD, A.D Nagar, Tripura, India.
- Online International Training on "Learning Science and impact of e-Learning on Rural Development Training" organized by the Panchayat Raj Training Institute, A. D. Nagar, Agartala on 24th & 25th August, 2020.
- Online Training of Trainers Programme on "Certification of Master Resource Persons" held from 31st August to 03rd September, 2020 organized by NIRDPR, Hyderabad, Government of India.
- Online Training of Trainers Programme on "Institutionalisation of Sustainable Development Goals (SDGs) through Panchayats to manage COVID-19 crisis" held from 15th September to 18th September, 2020 organized by NIRDPR, Hyderabad, Government of India.
- Online International Training on "Disaster Risk Resilience Leadership" organized by the Panchayat Raj Training Institute, A. D. Nagar, Agartala on 24th & 25th September, 2020.
- The International Webinar on Application of ICT in Higher Education During Covid-19 Pandemic Jointly Organized by IQAC & IGNOU Cell of Government Degree College, Kamalpur in collaboration with IGNOU, Agartala Regional Centre on 2nd September 2020.
- The Webinar on National Education Policy 2020 Organized by Bir Bikram Memorial Government Degree College, Agartala, Government of Tripura on 18th September 2020.
- Online Training of Trainers Programme on "Spatial Planning for GDPD" held from 21st September to 25th September, 2020 organized by NIRDPR-NERC, Guwahati, Government of India.
- Online Webinar Programme on "Applications of Open Source Technologies in Rural Development" held from 06th - 09th October, 2020, organized by at NIRD&PR-NERC, Guwahati, Government of India.
- Online Training of Trainers Programme on "Online ToT on Effective Utilization of 15th Finance Commission Grants" held from 12th October to 15th October, 2020 organized by NIRDPR, Hyderabad, Government of India.

- Online Orientation Training on "Panchayat Enterprise Suites (PES) and eGramSwaraj" held from 07th - 10th October, 2020, Certificate ID: XCV2UQ-CE000011 organized by at NIRD&PR-NERC;, Guwahati, Government of India.

2019

- "One week Trainer Development Programme on Experiential Learning Tools (ELT)" held from 15th - 19th January 2019 at SIPARD, A.D Nagar, Tripura, India.
- Training Programme on "ICT Applications in Office Automation" held from 12th - 16th February, 2019 at NIRD&PR-NERC;, Guwahati, Government of India.
- Presented Project Proposal (for Innovation) from Panchayat Department, Government of Tripura in the meeting on CEC Rashtriya Gram Swaraj Abhiyan (RGSA), Ministry of Panchayati Raj, Government of India held on 13th June, 2019 at Vigyan Bhawan, New Delhi.

2018

- Training of Trainers Programme on "e-Enablement of Gram Panchayats: PES and PFMS" held from 18th - 22nd June, 2018 at NIRDPR, Hyderabad, Government of India.
- "Comprehensive Online Modified Modules for Induction Training (COMMIT)" - workshop for Training of Trainers held on 11th October, 2018 at SIPARD, A.D Nagar, Tripura, India.

2017

- International Seminar on "Rural Habitat, Institutions and Development: Changing Nature & Challenges" held from 5th - 7th January, 2017 at Tripura University, India.
- National Level ToT Programme on "Enabling Digital Payments in Rural India" held from 17th - 18th January, 2017 at NIRD&PR-NERC;, Guwahati, Government of India.
- Training cum Sensitization Programme on "Rooftop Off Grid Solar Energy Technologies" held from 14th - 16th February, 2017 at Panchayat Raj Training Institution, Udaipur, Gomati District, India.
- "Training Course on Training Need Analysis" held from 30th October - 04th November, 2017 at SIPARD, A.D Nagar, Tripura, India.
- "Three Day Course on Facilitation Skills" held from 11th - 13th December, 2017 at SIPARD, A.D Nagar, Tripura, India.
- 2 Day District Level Training of Trainers on "Panchayat Enterprise Suite (PES) Applications" held on 27th & 28th December, 2017 at PRTI, A.D Nagar, Tripura, India.
- 48th Regional Science Conference on "Rural Habitat, Institutions and Development: Changing Nature & Challenges" held from 5th - 7th January, 2017 at Tripura University, India.

2016

- "One Month Induction Training Programme for the Panchayat Resource Development Officers (PRDOs)" held from 1st February - 28th February, 2016 at PRTI, A.D Nagar, Tripura, India.
- 7 Days Training Programme on "e-Panchayat Applications" held from 1st - 7th June, 2016 at PRTI, A.D Nagar, Tripura, India.
- Training Programme on "e-Governance Applications for RD in context to the Digital India" held from 20th - 24th June, 2016 at NIRD&PR-NERC;, Guwahati, Government of India.
- "Two Days Training Programme of Officer In-Charge PRTI (DPOs) & Panchayat Resource Development Officers (PRDOs)" held from 10th - 11th November, 2016 at PRTI, A.D Nagar, Tripura, India.
- "Induction Training Programme for the Cutting Edge Level Officials" held from 14th - 27th December, 2016 at SIPARD, A.D Nagar, Tripura, India.

2015

- Two days National Workshop on "Human Computer Interaction" held on 13th -14th February, 2015 at National Institute of Technology Agartala (NIT-A), India.
- Two days Faculty Development Programme on "Recent Trends in R.F. and Wireless Communication" held on 13th -14th March, 2015 at Tripura Institute of Technology, Narsingarh, India.

- One day Workshop on "Intel HPC Code Modernization (Parallelization)" held on 06th August, 2015 at National Institute of Technology Agartala (NIT-A), India.

2014

- "Specialized Training for e-Governance Program on Information Security Management" which was held during 18th - 20th November, 2014 at Agartala, organized by National Institute for Smart Government and Directorate of Information Technology, Govt. of Tripura under National eGovernance Plan.
- "2014 International Conference on Issues and Challenges in Intelligent Computing Techniques" held on 7th-8th February, 2014 at Krishna Institute of Engineering & Technology Ghaziabad (KIET), India.
- North-East ISI-NIT Agartala SPRING SCHOOL on "Algorithms for Mobile and Pervasive Computing" held on 24th-26th February, 2014 at National Institute of Technology Agartala (NIT-A), India.
- ISI-TU WINTER SCHOOL on "Pattern Recognition and Image Processing" held on 25th-29th March, 2014 at Tripura University (A Central University), India.
- One week Faculty Development Programme on "Advanced Software Engineering" held on 26th-30th September, 2014 at Tripura Institute of Technology, Narsingarh, India.
- Two days National Workshop on "Cloud Computing" held on 20th -21st October, 2014 at National Institute of Technology Agartala (NIT-A), India.

2013

- Two days National Workshop on "MATLAB and its Application in Digital Image Processing and Pattern Recognition" held on 29th-30th November, 2013 at National Institute of Technology Agartala (NIT-A), India.
- "Workshop on Image and Speech Processing (WISP 2013)" held on 13th-14th December, 2013 at Indian Institute of Technology Guwahati (IIT-G), India.

ACHIEVEMENTS

- Qualified Graduate Aptitude Test in Engineering (GATE) twice in the year 2012 and 2015.
- Qualified OPENMAT-XL (Management Programme) Test, IGNOU in the year 2016.
- Best Paper Award for the paper title 'Locating Optic Disk in Retinal Fundus Images using Normalized Cross Correlation' at ERCICA 2014, ELSEVIER, Bangalore, INDIA.
- Ranked 1st in the Entrance Examination for admission to M.Tech in CSE Department conducted by Tripura University (Central University) in the year of 2012.
- Consistency in increasing CGPA marks in every semester during M.Tech Curriculum in Computer Science and Engineering Department at NIT Agartala.
- Have participated in various school and district level painting competitions and won many 1st prizes.
- Scored consecutively 95 percentile in state level Aptitude Test conducted by NIIT in the year of 2010 and 2011.
- Excellent performance in course titled Basics of Computers, Internet, MS Office, RDBMS and C++ certified by NIIT in 2006.
- Cleared Tripura Public Service Commission Exam (TPSC) twice for Group - A & Group - B Post in the year 2015.
- Certified as Training Need Analysis (TNA) Consultant by joint venture of Training Division, Department of Personnel & Training, Government of India and SIPARD, Tripura on 04th November 2017.
- Certified as Faculty Member in 12 Day Dronacharya 0.1 Course on Basic Android App Development during 25th Dec, 2019 - 05th Jan, 2020 organized by NGO: SEBALOY, Agartala, Tripura.
- Coordinated and delivered 150 minutes session on Digital India and Transforming India on 06/07/2015. Gomati District awarded as Best Performing District in India to spread awareness about Digital India at grass-root levels during Digital India Week Campaign 2015.
- Designed and Developed the Website for Bir Bikram Memorial College, Government of Tripura in the year 2014 using .Net, Cascading Style Sheet and Java Script.
- Certified by Google for successful completion of The Fundamentals of Digital Marketing certification exam with Registered Certificate ID: ANB ZGJ RRR.

- Designed and Developed Tripura Calendar 2020: Android App and published in Google Play Store with 3500+ Downloads.
- Invited Speaker in Regional Meet on Geo-Spatial Technologies in 2018 at NIT Agartala.